TUTORIAL-4

x_1 Study Hours	x_2 Attendance	x_3 Assignment	x_4 Internal score	Results (y)	$\hat{y}_i$	$(y-\hat{y}_i)$	$(y-\hat{y})x_i$
12	85	80	35	1	0.5	0.5	
6	60	55	22	0	0.73	-0.73	
15	90	88	40	1	0.5	0.5	
5	58	50	20	0	0.73	-0.73	
10	75	70	30	1	0.5	0.5	
8	65	60	25	0	0.73	-0.73	
18	95	92	45	1	0.5	0.5	
7	62	58	24	0	0.73	-0.73	
14	88	85	38	1	0.5	0.5	

Find the equation for Logistic regression

Sol:

Sigmoid function.

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$z = w_0 + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4$$

$w_0 = 0,\ w_1 = 0,\ w_2 = 0,\ w_3 = 0,\ b = 0$

$$z_1 = 0 + 0(12) + 0(85) + 0(80) + 0(35)$$

$$z_1 = 0$$

$$\sigma(z_1) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-0}} = \frac{1}{2} = 0.5$$

$$\hat{y}_1 =$$

$$w_0^{new} = w_0^{old} - \frac{\eta}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)x_i$$

$$= 0 - (0.1)(0.5)(12)$$

$$w_1^{new} = -0.6$$

$$y \log \hat{y} + (1-y)\log(1-\hat{y})$$

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